Assignment

Subject: programing_fundamental_lab_5

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Section: D

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Date: 17 oct 2026

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Task1:

Write a c++ program to print table of any number.

```
#include<iostream>

using namespace std;

int main()

{

	int i,j;

	cout<<"Enter a number: ";

	cin>>i;

	for(j=1;j<=10;j++)

	{

		cout<<"\n";

		cout<<i<<"\t"<<j<<"\t"<<(i*j)<<endl;

	}

	return 0;

}
```

Output:

```
Enter a number: 10


10	1	10


10	2	20


10	3	30


10	4	40


10	5	50
```

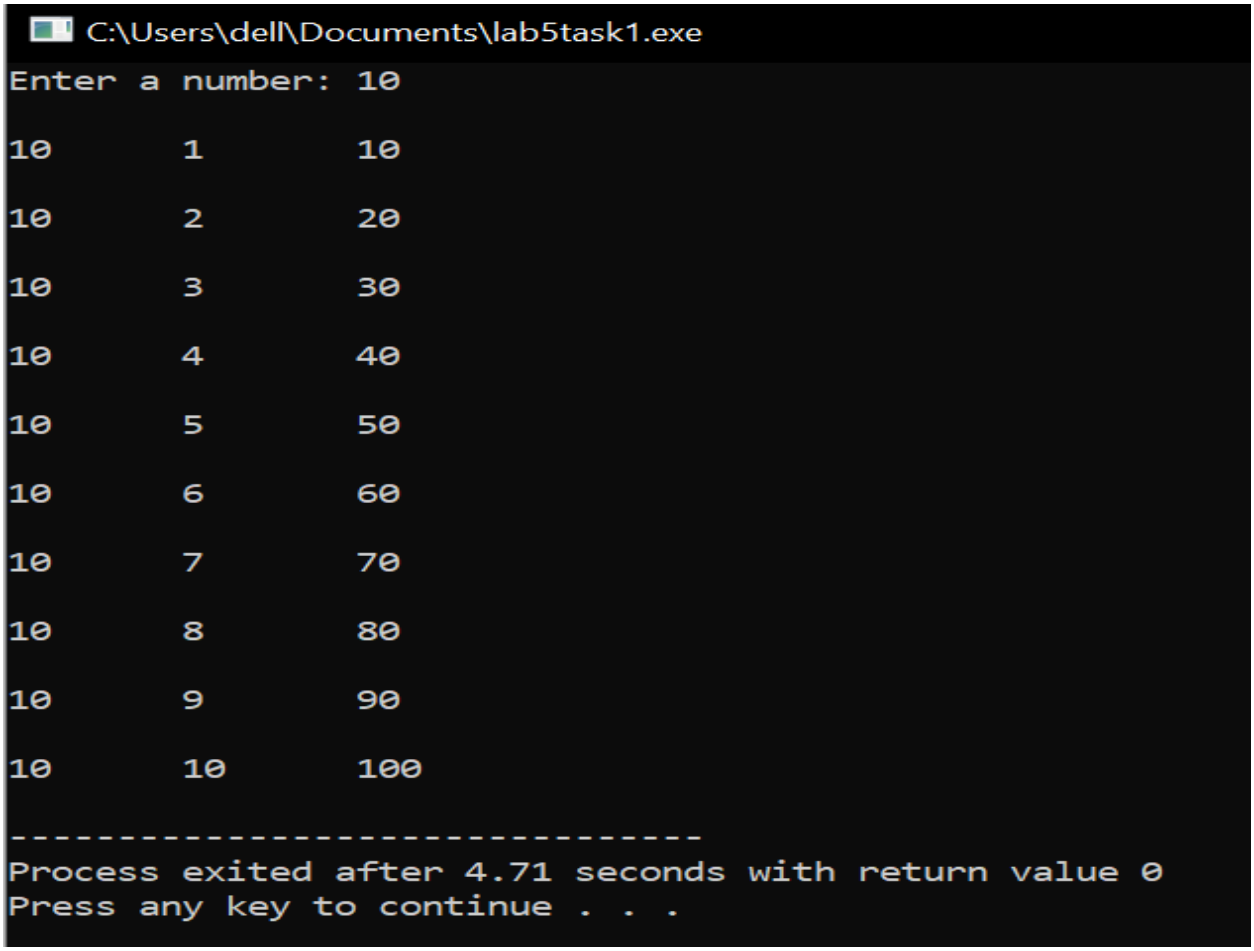
10 6 60

10 7 70

10 8 80

10 9 90

10 10 100



Task2:

Find the power of any number using for loop.

```
#include <iostream>

using namespace std;


int main() {

   int number, power, result = 1;


   cout << "Enter number and power: ";

   cin >> number >> power;


   for (int i = 0; i < power; i++) {

      result *= number;

   }


   cout << "Answer: " << result;

   return 0;

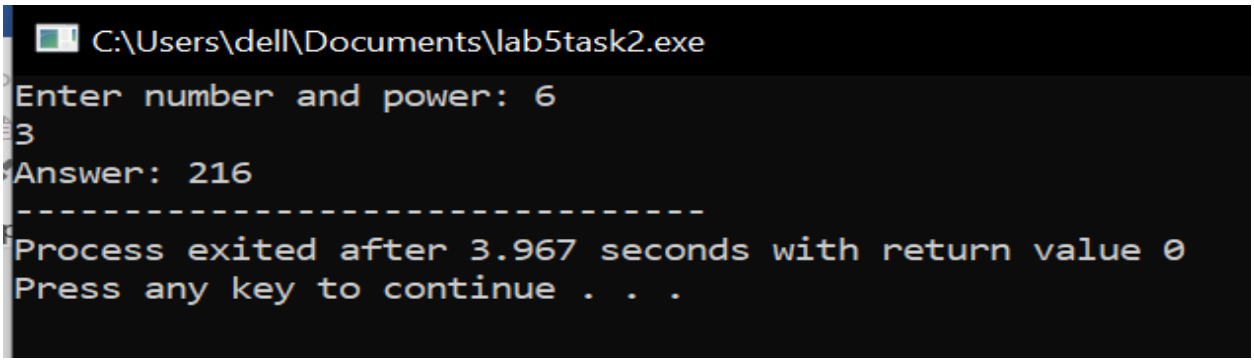
}
```

Output:

Enter number and power: 6

3

Answer: 216



Task3:

Sum of natural numbers using loop.

```
#include<iostream>

#include<math.h>

using namespace std;

int main()

{

	int i,sum=0;

	cout<<"Enter a number: ";

	cin>>i;

	for(int j=0;j<=i;j++)

	{

		sum += j;

	}

	cout<<"sum: "<<sum;

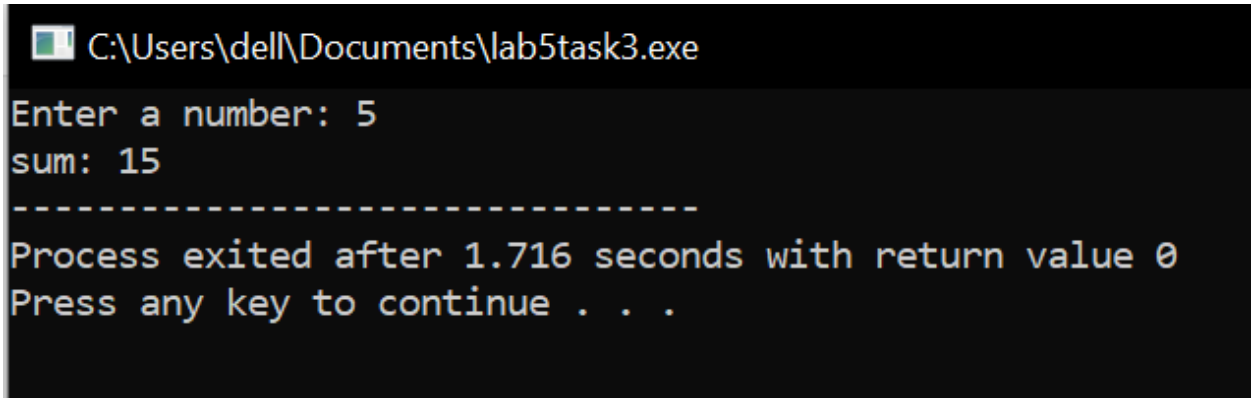
	return 0;

}
```

Output:

Enter a number: 5

sum: 15



Task4:

C++ program to find sum of digits of a number.

```
#include<iostream>

using namespace std;

int main()

{

	int i,j=0;

	cout<<"Enter number: ";

	cin>>i;

	for(;i!=0;i/=10)

	{

		j += i%10;

	}

	cout<<"sum of digits is: "<<j;

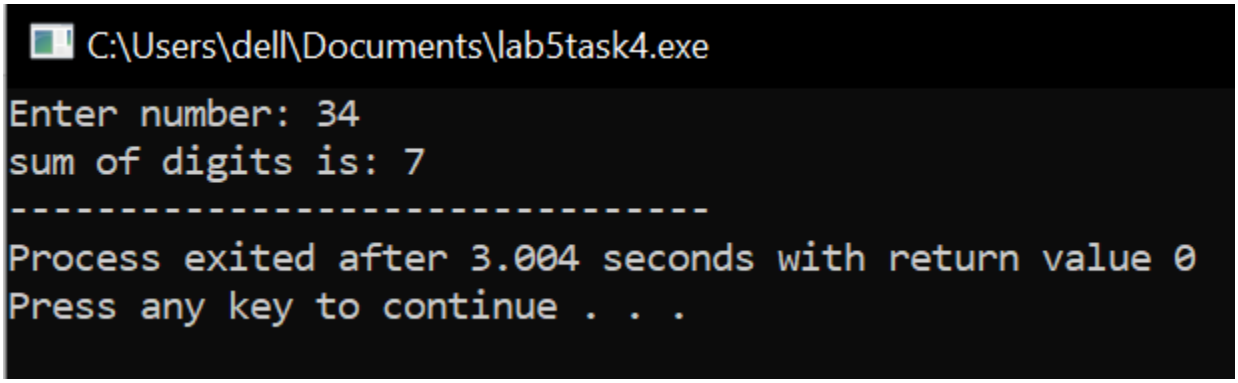
	return 0;

}
```

Output:

Enter number: 34

sum of digits is: 7



Task5:

Write a number to generate factorial a certain number.

```
#include<iostream>

using namespace std;

int main()

{

	int i,j=1;

	cout<<"Enter a number: ";

	cin>>i;

	for(; i != 0; i/=10)

	{

		j *= i%10;

	}

	cout<<"factorial of number is: "<<j;

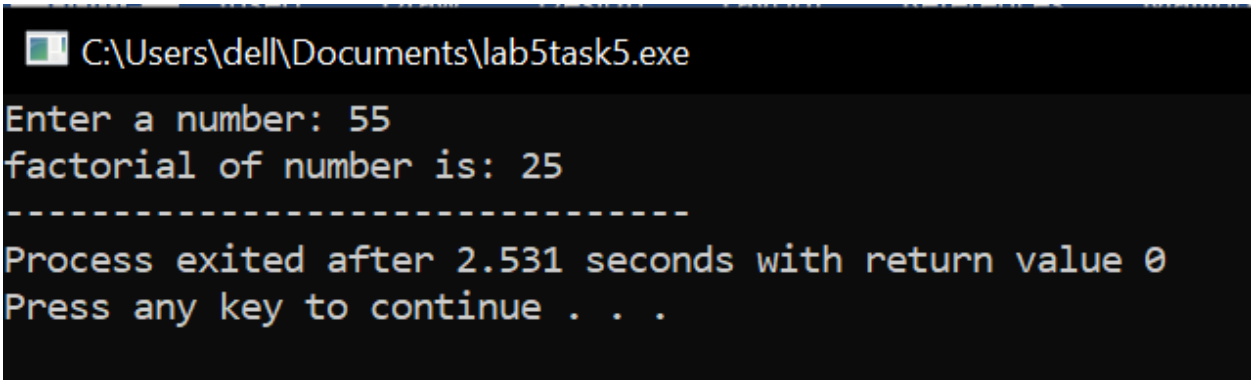
	return 0;

}
```

Output:

Enter a number: 55

factorial of number is: 25



Task6:

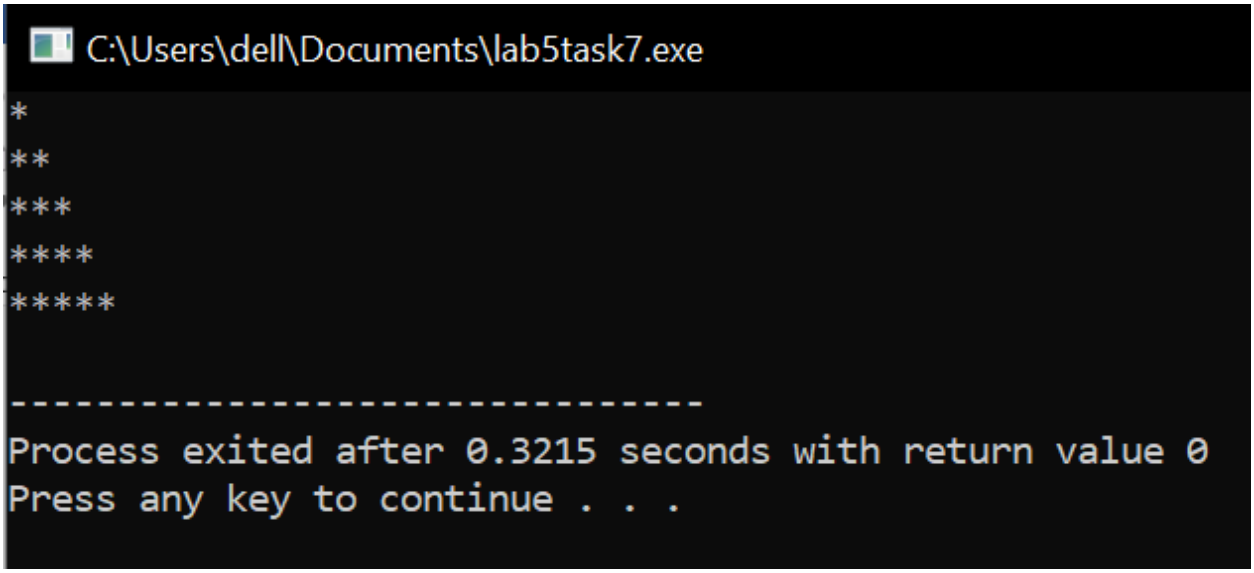
Write a c++ program that will write the following:

```
*
**
***
****
*****
```

```
#include<iostream>
using namespace std;
int main()
{
	int i,j;
	for(i = 1; i <= 5; i++)
	{
		for(j = 1; j <= i; j++)
		{
			cout<<"*";
		}
		cout<<endl;
	}
	return 0;
}
```

Output:

```
*
**
***
****
*****
```



Task7:

Write a c++ program print total number of days in a month using switch case.

```
#include<iostream>

using namespace std;

int main()

{

	int month;

	cout<<"Enter a month number: ";

	cin>>month;

	switch(month)

	{

		case 1: case 3: case 5: case 7: case 8: case 10: case 12:

		cout<<"total days in this month is: 31";

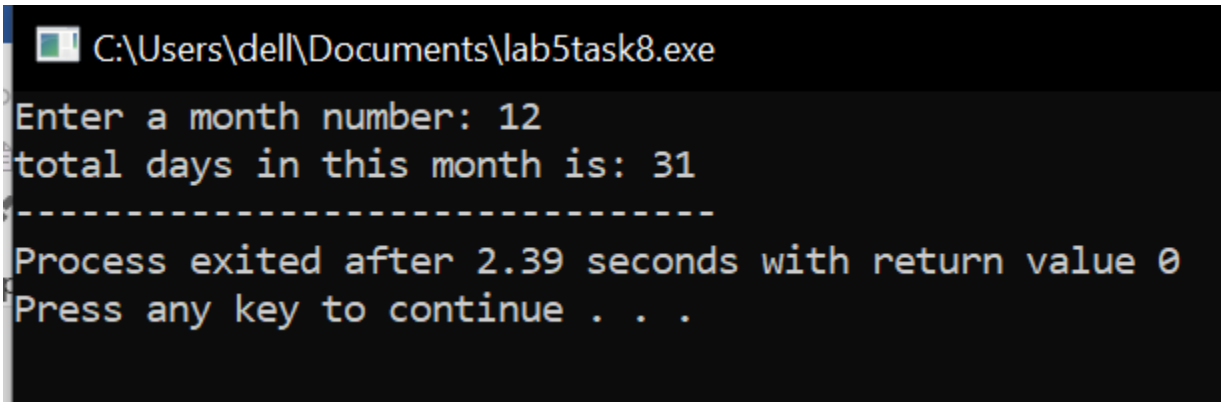
	   break;

	   case 4: case 6: case 9: case 11:
```

```
            cout<<"total days in this month is: 30";
            break;
            case 2:
            cout<<"total days in this month is: 28 or 29";
            break;
            defaul:
            cout<<"invalid month number try again";
    }
    return 0;
}
```

Output:

```
Enter a month number: 12
total days in this month is: 31
```



Task8:

Write a c++ program to check whether an alphabet is vowel or constant using switch case.

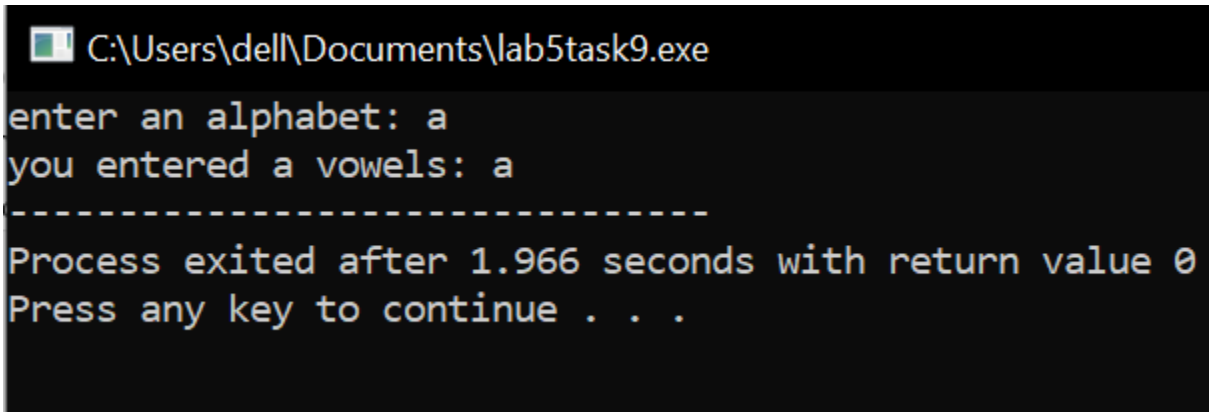
```
#include<iostream>
using namespace std;
int main()
{
        char b;
```

```
	cout<<"enter an alphabet: ";
	cin>>b;
	switch(b)
	{
		case 'a': case 'e': case 'i': case 'o': case 'u':
			cout<<"you entered a vowels: "<<b;
			break;
			case 'A': case 'E': case 'I': case 'O': case 'U':
				cout<<"you entered a vowel: "<<b;
			default:
				cout<<"you entered a charector: "<<b;
	}
	return 0;
}
```

Output:

enter an alphabet: a

you entered a vowels: a



Task9:

1.What will be the output of the following programs.

```
#include<iostream>

using namespace std;
```

```
int main()
{

	for(5;2;2)
	{
		cout<<"hello";
	}
	return 0;
}
```

Output:

It will be showing Infinite 'hello' on the output screen.

2.what will be the output of the c program, if input is 6?

```
#include<iostream>
using namespace std;
int main()
{
	int i;
	for(i = 0;i > 9;i+=6)
	{
		cout<<"for";
	}
	return 0;
}
```

Output:

It will be not print any thing because condition remains falls (i=0; i>9; i+=6)

3.what will be the output of the c++ program.

```
#include<iostream>
using namespace std;
```

```
int main()
{
	for(;;)
	{
		cout<<"10";
	}
	return 0;
}
```

Output:

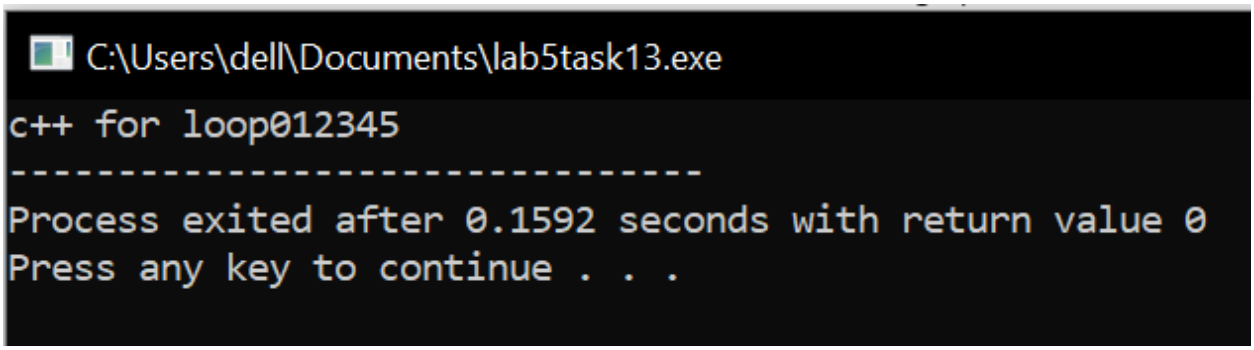
This program will be also show infinite 10 on the output screen.

4.what will be the output of the cprogram?

```
#include<iostream>
using namespace std;
int main()
{
	int fun=5;
	cout<<"c++ for loop";
	int x = 5;
	for(x = 0; x <= fun; x++)
	{
		cout<<x;
	}
	return 0;
}
```

Output:

c++ for loop012345


C:\Users\dell\Documents\lab5task13.exe
c++ for loop012345

Process exited after 0.1592 seconds with return value 0
Press any key to continue . . .